

Agriculture Price Prediction using Decision Tree Regressor & Time Series Analysis

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Abstract— The paper "Research Paper on Agriculture Price Prediction" examines the critical functions of the agricultural sector globally and highlights its role in producing the goods that preserve human life and support almost all other industries. Effective prediction of agricultural commodity price is presented as an essential and necessary need for farmers and policymakers alike and for consuming communities because it provides the informed decision-making capability, effective management of resources, and stabilization of prices. The reason it is complex is that the agricultural prices interconnect with several factors like climate conditions, market trends, and policy regulations, which is a bit challenging to determine the right price. With the complexity attached to the agricultural sector, issues about predicting the prices that are accurate arise because, most of the time, climate factor changes often characterize demand for agricultural commodities, bringing about considerable variability in the prices. Volatility in agricultural markets also makes it challenging to develop stable long-term forecasting models. Among the factors shaping agricultural production and related market dynamics, there are two broad categories: environmental ones - weather conditions and climate change - and economic ones, including demand from the market and policies from the government. Each category operates directly on prices for commodities. Since agriculture and the global economy are so inextricably linked, it is important that such predictions are made for agriculture price in order to guarantee economic stability and food security. Price volatility requires well-formulated models that can be used by stakeholders in understanding the intricacies of agricultural markets and making better-informed decisions.

Keywords— Machine Learning, Decision Tree Regressor, Future Forecasting.

I. INTRODUCTION

The agricultural sector is the backbone of the global economy, producing and providing essential goods that sustain human life and support other industries. An accurate forecast of agricultural commodity prices is therefore important for farmers, policymakers, and consumers alike. It helps make informed decisions, effectively allocate resources, and manage price volatility.

However, predicting agricultural prices is a complex task, as it involves the interplay of various factors, including climate, market dynamics, and policy regulations.

It is in this direction that reliable models for the prediction of agricultural prices have been developed, going from statistical analysis to machine learning approaches. This research paper aims to outline, as a whole, the current state of art in agricultural price prediction together with all challenges, methodologies, and applications these models may involve.

A. Challenges in Agricultural Price Prediction

The inherent complexities of the agricultural sector impose various difficulties in the formation of precise price predictions. For example, the agricultural products are usually in higher demand, which is strongly dependent on climate factors; such climate factors are relatively unpredictable and subject to sharp fluctuations. Moreover, large price volatilities characterise the agricultural market. Thus, it is impossible to construct reliable long-run forecasting models. This complexity is furthered by a myriad of variables that include soil characteristics, crop rotation, weather patterns, water availability, nitrogen levels, surface temperature, and a myriad of other economic factors in the dynamics of agricultural production and markets.

B. Factors Influencing Agricultural Prices

Agricultural price prediction is a hard task since various factors play their roles in changing commodity prices. On that account, they can be considered to some extent as falling into environmental and economic factors. Negative environmental factors, such as weather patterns, climate change, and other natural disasters, impact agriculture production, affecting the commodity prices.

Precipitation, temperature, and soil conditions contribute much to determining crop yields and hence affect the supply and demand factors in the market. This may involve other economic factors; including market demand, international trade, government policies and indicators of macroeconomic developments-all of which determine a shift in supply and demand to drive agricultural price inflation.

C. Literature Survey

This research will focus on high-precision machine learning algorithms such as Decision Trees and Random Forest for the prediction of agricultural prices for better accuracy and real-time deployment. Some supporting studies present various approaches in this area:

- Jha and Sinha (2013) investigated the prediction of wheat prices using the ARIMA model, which showed limitations in adapting to sudden price fluctuations, indicating a need for non-linear models in volatile markets. This highlights the necessity for more flexible modelling techniques in agricultural price forecasting.
- Bodnar et al. (2020) aimed to predict corn prices using machine learning algorithms, specifically Random Forest and SVM. Their findings demonstrated that these machine-learning models outperformed traditional statistical methods, suggesting that such approaches can significantly enhance the reliability of agricultural price forecasting.
- Sarker, et al. (2019) introduced a Hybrid Model of integrating ANN with the GA to forecast the rice price in the Indian market. An integration of the rainfall variable with history data related to the prices results in high accuracy levels. Conclusion In the multiplicity of sources and algorithms, there exists a better performance while producing forecasting performance.
- Al-Masri et al. (2019) examined the use of Decision Tree algorithms for predicting vegetable prices and found that this method achieved higher accuracy in volatile markets compared regression to conventional methods. Their research underscores the effectiveness of machine learning techniques, particularly scenarios with high price variability.
- Patil and Deshmukh (2021) developed a web-based system using Flask to predict sugarcane prices in real time. This system enhances accessibility for farmers and stakeholders, providing timely information that can aid in informed decision-making. Such technology-driven solutions can empower farmers by delivering valuable market insights and improving their operational strategies.

Collectively, these studies underscore the significance of innovative modelling techniques and real-time solutions in agricultural price forecasting, aiming to empower stakeholders with accurate market insights.

D. Methodologies for Agricultural Price Prediction

This study aims to predict the prices of general agricultural commodities in India using a combination of Random Forest and Time Series algorithms. The research will leverage historical data of commodity prices, incorporating relevant environmental and economic factors to develop a robust prediction model.

Data Collection

The study will utilize secondary data encompassing a significant timeframe (at least 10 years) to capture long-term trends and seasonality. The data will be collected from reputable sources such as:

- Price Data: Ministry of Agriculture & Farmers Welfare, India o Commodity Exchanges o Agricultural Marketing Information Network

- Environmental Data: o India Meteorological Department for rainfall, temperature, and other weather variables.
- Economic Data: o Reserve Bank of India for inflation, interest rates, and other macroeconomic indicators. o Ministry of Commerce & Industry for import/export data.

Data Pre-processing

The data collected will undergo preliminary processing to ensure the quality and appropriateness for further analysis. Such steps include:

- Data cleaning: Imputed or deleted missing values when the amount of missing data was considerable.
- Data Transformation: transform data to an appropriate scale for analysis purpose, for example, using log-scale to handle skewness in price data.
- Feature Engineering: Creating additional features from existing ones that help the model increase accuracy further. Examples include lagging price variables or moving averages.
- Data Splitting: Split the data into 80-20%, and allocate first part of it for training and the rest for testing.

Model Development

The study will adopt two altogether different yet complementary approaches to price prediction:

- Decision tree: The Random Forest algorithm, being a robust ensemble learning method, can capture the non-linear relationship between predictor variables and the target variable (consumer prices).
- Time Series Analysis: Techniques like ARIMA or SARIMA will be employed to model the temporal dependencies and seasonality inherent commodity prices.

Model Evaluation

The performance of both models will be evaluated based on standard metrics:

- Root Mean Squared Error: Measures the overall accuracy of the predictions.
- Mean Absolute Percentage Error: This measure provides results which have interpretable meaning as a measure of percentage accuracy in predicting.
- R-squared (R^2): The percentage variance of the dependent variable accounted for by the model.

Model Comparison and Selection

The performance of the Decision Tree and Time Series models will be compared based on the evaluation metrics. The model with the best performance, indicating higher accuracy and better predictive power, will be selected as the preferred model for agricultural commodity price prediction in India.

Limitations

This research also identifies some limitations to consider:

- Availability of Data: Accuracy of models is highly dependent on the quality and completeness of collected data.
- Assumption of Models: Assumptions are imbibed both in the Decision Tree and Time-Series models, and these may not always be true.
- Shock External Factors: Events such as disease outbreaks or changes in government policies that occur unexpectedly can disrupt the markets for agriculture and also restrict the forecasting ability of the models.

System Design

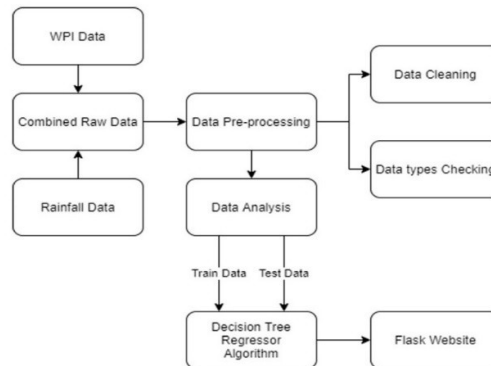


Fig.1 Architecture Diagram

The following flowchart in our study outlines the process that would be utilized to predict agricultural prices. To initiate the agricultural price prediction process, there are two main data inputs: WPI (Wholesale Price Index) Data concerning the variation of wholesale price and Rainfall Data due to its great effect on the agricultural output. Analysis is further conducted from those two combined into a singular raw dataset. Data will first need to be pre-processed order to correct inaccuracies and/or inconsistencies; therefore, check data types in order to get all variables in proper formats for analyses.

The processed dataset was subjected to analysis to derive trends and patterns. The data was divided into two portions, namely, the training data for training the predictive model and the testing data to evaluate the model accuracy. A Decision Tree Regressor Algorithm is used to predict the future prices of agriculture commodities based on the input variables like rainfall and WPI. Once the model produces the prediction, results are deployed on a website made on Flask where it allows the farmers or any decision-making people to log in and find out what prices will be for particular commodities. The whole procedure relies on machine learning as well as data analysis.

Results

```

model = DecisionTreeRegressor(random_state=42)
model.fit(X_train, y_train)
y_pred = model.predict(X_test)

mae = mean_absolute_error(y_test, y_pred)
mse = mean_squared_error(y_test, y_pred)
rmse = rmse_score(y_test, y_pred)
r2 = r2_score(y_test, y_pred)

print("Mean Absolute Error (MAE):", mae)
print("Mean Squared Error (MSE):", mse)
print("Root Mean Squared Error (RMSE):", rmse)
print("R-squared score (R2):", r2)

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Fig.2 Performance Indicators

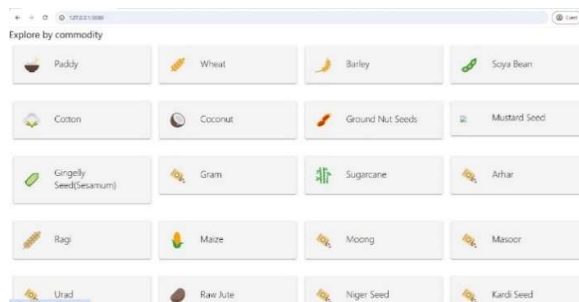


Fig. 3 Choice of Crops

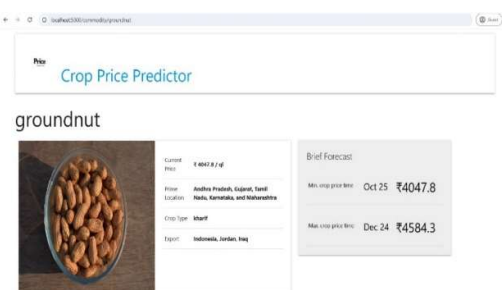


Fig. 4 Output for Groundnut

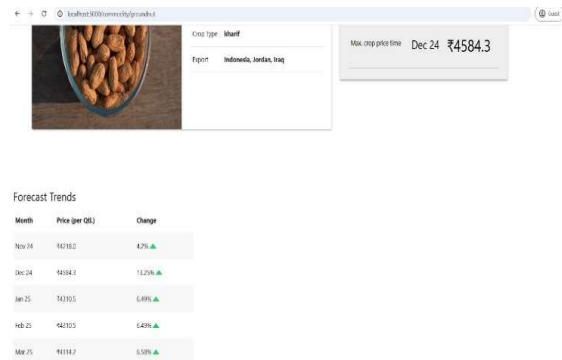


Fig.5 Output for Groundnut

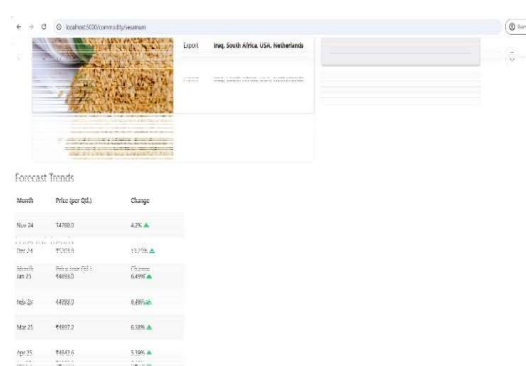


Fig. 6 Output for Sesamum

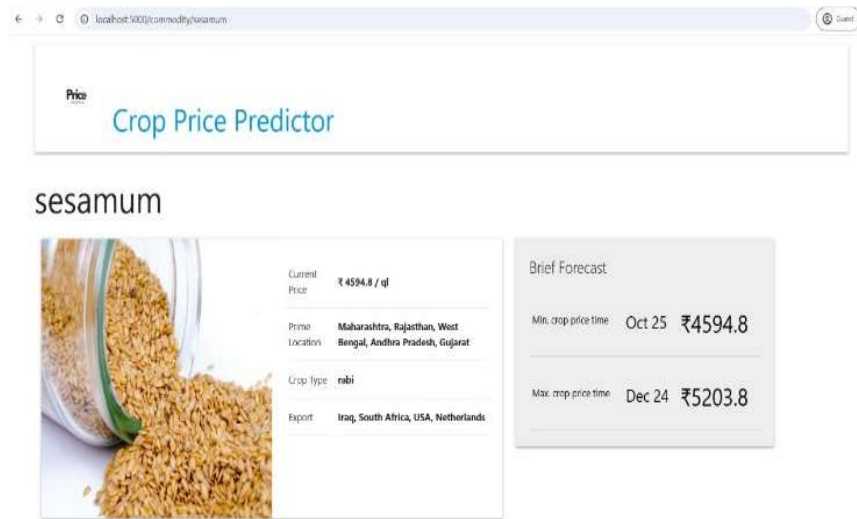


Fig. 7 Output for Sesamum

II. DISCUSSION

- It is one of the most critical sectors of the world economy, and price prediction is essential for all its participants, be it farmers, policymakers, or consumers. The paper states that reliable models in agricultural price prediction can ease informed decision-making, effective resource allocation, and crucially alleviate price volatility and food security issues.
- Agricultural price forecasting is difficult due to several reasons like climatic, market dynamics, and policy regulations. As per the paper, environmental factors, such as weather and climate change, play a crucial role in agricultural production and thus commodity prices. Other studies have also reflected that climate variability causes crop yields to swing by significant amounts, hence making the price forecasting tough.
- Classifies the available agricultural price prediction models into traditional, intelligent, and hybrid methods. This study has recognized traditional methods, with applications usually statistical analysis, dependent on while intelligent approaches have utilized machine learning techniques. The review shows that hybrid models may exhibit superior accuracy and reliability by combining traditional and advanced approaches. Other researchers have also indicated that in some scenarios, hybrid models would outperform single method models.
- Data quality and availability are critical challenges in developing accurate prediction models. The paper emphasizes that incomplete or poor-quality data can lead to inaccurate predictions, highlighting the importance of robust data collection processes. This concern is echoed in other studies, which indicate that the lack of reliable data can severely hinder the effectiveness of predictive models in agriculture. There is potential for overcoming some of the burdens of price volatility in the agricultural sector by offering new approaches such as machine learning and artificial intelligence. Even though these technologies bring better clarity to the market, resource allocation enhancement, and decision-making, they go on to enhance a sustainable agriculture sector.
- However, the paper recognizes that agricultural price prediction models have significant limitations. External shocks, such as pandemics or sudden policy changes, can disrupt agricultural markets and limit the models' predictive power. This unpredictability underscores the need for models that can adapt to changing conditions and incorporate uncertainties. Other literature emphasizes the importance of developing robust models that can withstand such shocks to maintain their relevance in real-world applications.
- The paper calls for further research to develop more accurate and reliable prediction models. This includes exploring new methodologies, improving data collection processes, and enhancing model robustness to external shocks. This is where the transformative power of AI in agricultural practices is highlighted, as current technological advancements could significantly enhance the accuracy of price prediction. Others have proposed that agricultural science, economics, and data science should merge to create innovation in this sector.

- The right decisions about resource use allocation, and risk management of stakeholders will be made due to proper agricultural price forecasting. Thus, investments in the agriculture sector through development or research are critical to ensuring economic stability and food security. Agricultural markets around the world are integrated; hence, interindustry or multidisciplinary work must be done to provide valuable insights into effectively managing the multidimensional role of agricultural price prediction.

III. CONCLUSION

To sum up, a research paper on Agriculture Price Prediction indicates the scope of the studies and the range of barriers within this domain. Given the complexities associated with agricultural markets, there is a need to improve existing prediction models to help diagnosed farmers, policymakers, and consumers among others. The combination of techniques such as Random Forest and Time Series analysis helps increase the level of prediction and this level of optimization assists resource management among the operators involved in the process. The paper also raises the issue that existing models have their drawbacks such as external shocks: like Covid 19 and policy dynamics that bear directly on the markets and their predictive ability is cut short. In addition to that, the underlying assumptions for both Random Forest and Time Series models are not necessarily valid outside the theoretical framework which shows the need for further development of the approaches.

The systematic literature review conducted within the paper suggests that agricultural price prediction has received increased attention over time and existing models in this area fall into three categories; traditional, intelligent and hybrid. This calls for almost an urgent need to do more research in this area in order to create better, more dependable and precise prediction models.

Furthermore, the future prospects of artificial intelligence and machine learning in positively transforming the programs of agriculture are highlighted. Special emphasis is put on how much they enhance the transparency of the markets and lead to better decision-making. Since agriculture still grapples with such issues as price fluctuations and adverse climatic effects, the development of impending, accurate prediction models will assume paramount importance in providing food security and economic security both. Overall, this paper critically reviews the developments concerning agricultural price forecasting and draws attention to the need for improvement in the existing ones, as well as the complex issues affecting agricultural prices. In addition, the results underline the need for concerted action by researchers, practitioners, and policy-makers in order to achieve the goals of a more productive agricultural sector despite the emerging challenges.

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